

PINUS

P. densiflora (Japanese red pine). Broadly conical to rounded conifer, becoming flat-topped. H 50–80ft (15–25m), S 15–22ft (5–7m). Has scaling, reddish-brown bark, slender, bright green leaves and conical, yellow or pale brown cones. Z3–7. ‘**Alice Verkade**’, H 20in (50cm) or more, S 3ft (1m), is a diminutive, rounded form with fresh green leaves. ‘**Tagyosho**’ see *P.d.* ‘*Umbraculifera*’. ‘**Umbraculifera**’ (syn. *P.d.* ‘*Tagyosho*’), H 12ft (4m), S 20ft (6m), is a slow-growing, rounded or umbrella-shaped form.

P. excelsa. See *P. wallichiana*.

P. griffithii. See *P. wallichiana*.

P. halepensis (Aleppo pine). Conical, open-crowned conifer. H 50–80ft (15–25m) S 22–40ft (7–12m). Has an open habit and bright green leaves, 2½–4½in (6–11cm) long, and ovoid, glossy, brown cones. Young trees retain glaucous, juvenile needles for several years. Z9–10

P. heldreichii, syn. *P. leucodermis* (Bosnian pine), illus. p.98. ‘**Compact Gem**’ is a broadly conical, dense, dwarf conifer. H and S 10–12in (25–30cm). Has scaly, ash-gray bark and very dark green leaves in pairs. Grows only 1in (2.5cm) a year. Z3–8. ‘**Smidtii**’ (illus. p.105) is a dwarf form with an ovoid habit and sharp, dark green leaves.

P. x holfordiana (Holford pine) illus. p.95.

P. insignis. See *P. radiata*.

P. jeffreyi (Black pine, Jeffrey pine) illus. p.97.

P. leucodermis. See *P. heldreichii*.

P. mugo (Dwarf mountain pine). Spreading, shrubby conifer or spreading tree. H to 11½ft (3.5m), S to 15ft (5m). Has scaly, gray bark, bright to dark green leaves in pairs and ovoid, brown cones. Z3–7. ‘**Gnom**’, H and S to 6ft (2m), and ‘**Mops**’ (illus. p.105), H 3ft (1m), S 6ft (2m), are rounded shrubs.

P. muricata (Bishop pine) illus. p.97.

P. nigra (Austrian pine, European black pine). Upright, later spreading conifer, generally grown in one of the following forms. Z4–7. ‘**Hornibrookiana**’, H 5–6ft (1.5–2m), S 6ft (2m), is shrubby with stout, spreading or erect branches and dark green leaves in pairs. **subsp. laricio** (syn. *P.n.* subsp. *maritima*; Corsican pine), H to 130ft (40m), S to 30ft (10m), is fast-growing and narrowly conical with an open crown; bears gray-green leaves, in pairs, and ovoid to conical, yellow- or pale gray-brown cones. **subsp. maritima** see *P.n.* subsp. *laricio*. **subsp. nigra** (Austrian pine) illus. p.98.

P. parviflora (Japanese white pine) illus. p.99. ‘**Adcock’s Dwarf**’ is a slow-growing, rounded, dense, dwarf conifer. H 6–10ft (2–3m), S 5–6ft (1.5–2m). Bears short, gray-green leaves in 5s. Z5–7

P. patula (Mexican weeping pine) illus. p.97.

P. peuce (Macedonian pine) illus. p.95.

P. pinaster (Cluster pine, Maritime pine) illus. p.97.

P. pinea (Stone pine, Umbrella pine) illus. p.104.

P. ponderosa (Western yellow pine) illus. p.97.

P. pumila (Dwarf Siberian pine). Spreading, shrubby conifer. H and S 6–20ft (2–6m). Has bright blue-green leaves in 5s. Ovoid cones are violet-purple, ripening to

red-brown or yellow-brown, the males bright red-purple in spring. Z4–8. ‘**Globe**’, H and S 1½–3ft (50cm–1m), is a rounded cultivar with blue foliage.

P. radiata, syn. *P. insignis* (Monterey pine), illus. p.98.

P. rigida (Northern pitch pine) illus. p.100.

P. strobus (Eastern white pine, Weymouth pine) illus. p.96.

P. sylvestris (Scots pine; illus. p.78). Conifer, upright and with whorled branches when young, that develops a spreading, rounded crown with age. H 50–100ft (15–30m), S 20–28ft (6–9m). Bark is flaking and red-brown on upper trunk, fissured and purple-gray at base. Has blue-green leaves in 2s and conical, green cones that ripen to pale gray- or red-brown. Z3–8. ‘**Aurea**’ illus. p.104. ‘**Beuvronensis**’, H and S 3ft (1m), is a rounded shrub. ‘**Doone Valley**’, H and S 3ft (1m), is an upright, irregularly shaped shrub. **f. fastigiata** see Fastigiata Group.

Fastigiata Group (syn. *P.s. f. fastigiata*), H up to 25ft (8m), S 3–10ft (1–3m), has a narrow, obelisk habit; suffers wind damage in an exposed site. ‘**Gold Coin**’, H and S 6ft (2m), is a dwarf version of *P.s.* ‘*Aurea*’. ‘**Nana**’ see *P.s.* ‘*Watereri*’. ‘**Watereri**’ (syn. *P.s.* ‘*Nana*’), H 12ft (4m), S 22ft (7m), is a very dense cultivar with widely spaced leaves.

P. thunbergii (Japanese black pine) illus. p.100.

P. wallichiana, syn. *P. chylla*, *P. excelsa*, *P. griffithii* (Bhutan pine, Himalayan pine), illus. p.97.

PIPTANTHUS

PAPILIONACEAE/LEGUMINOSAE

Genus of deciduous or semievergreen shrubs, grown for their foliage and pealike flowers. Frost hardy. In cold areas needs the protection of a warm, sunny wall. Requires sun and fertile, well-drained soil. In spring, cut some older shoots back to ground level and prune any frost-damaged growths back to healthy wood. Propagate by seed in fall.

P. laburnifolius. See *P. nepalensis*.

P. nepalensis, syn. *P. laburnifolius*, illus. p.206.

PISONIA

NYCTAGINACEAE

Genus of evergreen shrubs, trees, and climbers grown for their attractive foliage and overall appearance. Frost tender. Needs full light or partial shade and humus-rich, well-drained soil. Water containerized specimens freely when in full growth, moderately at other times. Pruning is tolerated if required. Propagate by seed in spring or by semiripe cuttings in summer.

P. brunoniana. See *P. umbellifera*.

P. umbellifera, syn. *Heimerliodendron brunonianum*, *P. brunoniana* (Bird-catcher tree, Para para). Evergreen, rounded large shrub or small tree, usually freely branching. H 15–70ft (5–20m), S 10–15ft (3–5m). Has elliptic to lance-shaped, leathery, glossy, rich green leaves. In spring, bears clusters of tiny, green or pink flowers, followed by 5-winged, sticky, brownish fruits. Z11–12 (min. 50°F/10°C)

PISTACIA

Pistachio

ANACARDIACEAE

Genus of evergreen or deciduous trees, grown for their foliage and overall appearance. Frost hardy to half hardy. Requires full light and free-draining, even dry soil. Water containerized plants moderately when in full growth, sparingly at other times. Pruning is tolerated if necessary. Propagate by seed in spring or by semiripe cuttings in summer.

P. lentiscus (Mastic tree). Evergreen, irregularly rounded shrub or tree. H and S 3–10ft (1–3m). Bears leaves that are divided into 2–5 pairs of ovate, leathery, glossy leaflets. Produces axillary clusters of insignificant flowers from spring to early summer that develop into round, red then black fruits in fall. Z10–11

P. terebinthus (Terebinth, Turpentine tree). Deciduous, rounded to ovoid tree. H 20–28ft (6–9m), S 10–20ft (3–6m) Leaves have 3–6 pairs of ovate, usually lustrous, rich green leaflets. Axillary clusters of small flowers, borne in spring and early summer, develop into tiny, round to ovoid, red then purple-brown fruits in fall. Z9–11

PITCAIRNIA

BROMELIACEAE

Genus of very variable, usually evergreen, rosette-forming perennials (bromeliads), grown for their overall appearance. Frost hardy to frost tender. Needs partial shade and well-drained soil. Water moderately in growing season, sparingly at other times. Propagate by offsets or division in spring.

P. andreana. Evergreen, clump-forming, basal-rosetted perennial. H 8in (20cm), S 1ft (30cm) or more. Loose rosettes comprise narrowly lance-shaped, strongly arching, green leaves, gray-scaled beneath. Racemes of tubular, orange-and-red flowers are borne in summer. Z11–12 (min. 50°F/10°C)

P. heterophylla. Evergreen, epiphytic or terrestrial, perennial with loose, basal rosettes and swollen, much-branched rhizomes. H 4in (10cm) or more, S to 1ft (30cm). Outer leaves resemble barbed spines, inner leaves are strap-shaped, low-arching and green, with downy, white undersides. Bears almost stemless spikes of tubular, bright red, or rarely white flowers, amid red bracts, in summer. Z10–11

PITTOSPORUM

PITTOSPORACEAE

Genus of evergreen trees and shrubs, grown for their ornamental foliage and fragrant flowers. Frost hardy to frost tender. Does best in mild areas; in cold regions, grow against a warm, sunny wall. *P. crassifolium* can be grown as a wind-resistant hedge in mild, coastal areas; forms with variegated or purple leaves prefer sun. Others will grow in sun or partial shade. All need well-drained soil. Propagate *P. dallii* by budding in summer, other species by seed in fall or spring or by semiripe cuttings in summer; propagate selected forms by semiripe cuttings only in summer.

P. ‘Arundel Green’, syn. *P. tenuifolium* ‘*Arundel Green*’. Evergreen, rounded shrub. H and S 5ft (1.5). Produces ovate, wavy-margined, bright green leaves on blackish stems. In late spring, scented, dark red flowers are borne in the leaf axils of mature plants. Z9–11

P. crassifolium (Karo). Evergreen, bushy-headed, dense, small tree or shrub. H 15–30ft (5–10m), S 6–15ft (2–5m). Has oblong, dark green leaves, gray-felted beneath. Clusters of small, fragrant, star-shaped, dark reddish-purple flowers are borne in spring. Z9–11

P. dallii illus. p.119.

P. eugenioides (Tarata). Evergreen, columnar tree. H 30ft (10m), S 6–15ft (2–5m). Narrowly ovate, wavy-edged leaves are glossy, dark green, and lemon-scented when crushed. Honey-scented, star-shaped, pale yellow flowers are produced in summer. Z9–11.

‘Variegatum’ illus. p.93.

P. ‘Garnettii’ illus. p.119.

P. tenuifolium illus. p.120. ‘**Arundel Green**’ see *P. ‘Arundel Green’*. ‘**Irene Paterson**’ is a slow-growing, large, bushy shrub or small tree. H to 4ft (1.2m), S to 2ft (60cm). Has glossy, usually wavy-margined, white leaves, speckled and mottled between the veins, on dark stems. Bears bell-shaped, honey-scented, blackish-red flowers in late spring to early summer. Z9–10. ‘**Tom Thumb**’ illus. p.164. ‘**Warnham Gold**’ is fast-growing, with wavy-edged, golden-green leaves that mature to golden-yellow.

P. tobira (Japanese mock orange). Evergreen, bushy-headed, dense tree or shrub. H 20ft (6m), S 12ft (4m). Has oblong to ovate, glossy, dark green leaves. Very fragrant, star-shaped, white flowers, opening in late spring, later become creamy-yellow. Z9–11. ‘**Variegatum**’ has gray-green leaves, irregularly margined white.

Plagianthus lyallii. See *Hoheria lyallii*.
Plagiorhegma dubia. See *Jeffersonia dubia*.

PLANTAGO

Plantain

PLANTAGINACEAE

Genus of summer-flowering annuals, biennials, and evergreen perennials and shrubs. Many species are invasive weeds, but a few are grown for their architectural value and foliage. Fully hardy to frost hardy. Needs full sun and well-drained soil. Water potted plants moderately, sparingly in winter. Propagate by seed or division in spring.

P. major. Variable, sometimes invasive perennial. H 8–16in (20–40cm), S to 1ft (30cm). Has rosettes of ovate, mid-green leaves. In summer, bears spikes of mid-green flowers on leafless stems. Z3–12. ‘**Atropurpurea**’ see *P.m.* ‘*Rubrifolia*’. ‘**Bowles’s Variety**’ see *P.m.* ‘*Rosularis*’. ‘**Rosularis**’, syn. ‘*Bowles’s Variety*’, is a mutation in which the flower spike develops as a cabbagelike rosette of small, ovate leaves. ‘**Rubrifolia**’, syn. ‘*Atropurpurea*’, has purplish-maroon leaves and is prone to mildew.

P. nivalis illus. p.376.